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Chapter #1: Signals and Amplifiers

from **Microelectronic Circuits** Text
by Sedra and Smith
Oxford Publishing

Introduction

- **IN THIS CHAPTER YOU WILL LEARN…**
  - That electronic circuits process signals, and thus understanding electrical signals is essential to appreciating the material in this book.
  - The Thevenin and Norton representations of signal sources.
  - The representation of a signal as sum of sine waves.
  - The analog and digital representations of a signal.

Introduction

- **IN THIS CHAPTER YOU WILL LEARN…**
  - The most basic and pervasive signal-processing function: signal amplification, and correspondingly, the signal amplifier.
  - How amplifiers are characterized (modeled) as circuit building blocks independent of their internal circuitry.
  - How the frequency response of an amplifier is measured, and how it is calculated, especially in the simple but common case of a single-time-constant (STC) type response.

1.1. **Signals**

- **signal** – contains information
  - e.g. voice of radio announcer reading the news
- **process** – an operation which allows an observer to understand this information from a signal
  - generally done electrically
- **transducer** – device which converts signal from non-electrical to electrical form
  - e.g. microphone (sound to electrical)

1.1: **Signals**

- **Q:** How are signals represented?
  - **A: thevenin form** – voltage source $\mathbf{v}_s(t)$ with series resistance R_S
    - preferable when R_S is low
  - **A: norton form** – current source $\mathbf{i}_s(t)$ with parallel resistance R_S
    - preferable when R_S is high

1.1. **Signals**

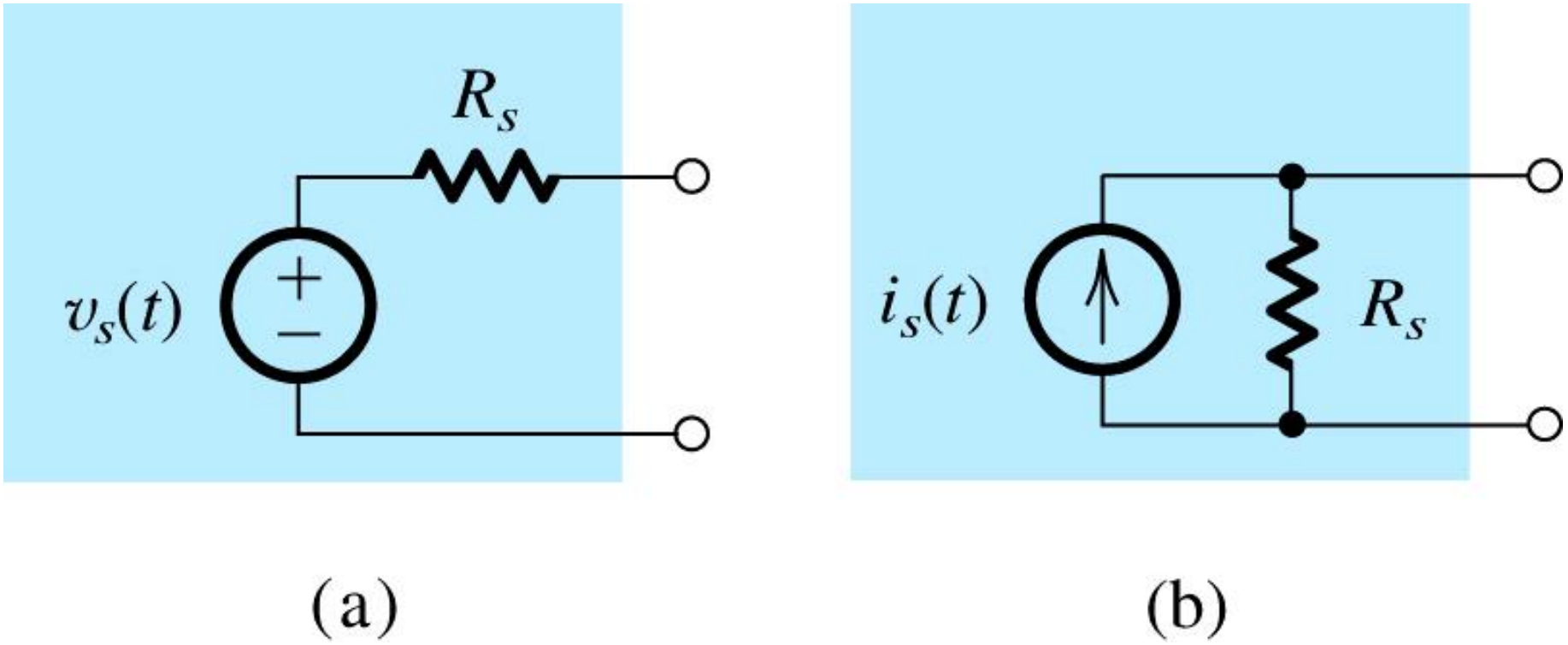
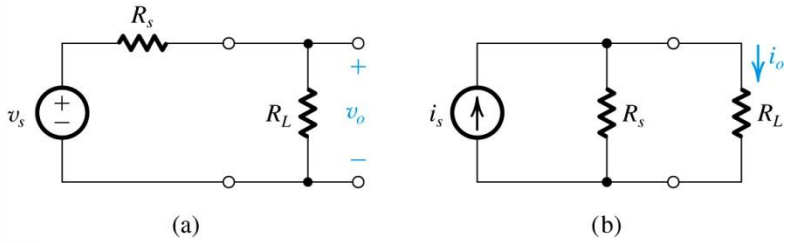


Figure 1.1: Two alternative representations of a signal source: **(a)** the Thévenin form; **(b)** the Norton form.

Example 1.1:
Thevenin and Norton Equivalent Sources



- Consider two source / load combinations to upper-right.
  - note that output resistance of a source limits its ability to deliver a signal at full strength
- **Q(a):** what is the relationship between the source and output when maximum power is delivered?
  - for example, $v_s < v_o$??? $v_s > v_o$??? $v_s = v_o$???
- **Q(b):** what are ideal values of R_S for norton and thevenin representations?

1.2. **Frequency Spectrum of Signals**

- **frequency spectrum** – defines the a time-domain signal in terms of the strength of harmonic components
  - **Q:** What is a Fourier Series?
    - **A:** An expression of a periodic function as the sum of an infinite number of sinusoids whose frequencies are harmonically related

What is a Fourier Series?

- **decomposition** – of a periodic function into the (possibly infinite) sum of simpler oscillating functions

Fourier Series Representation of $\mathbf{f}(x)$

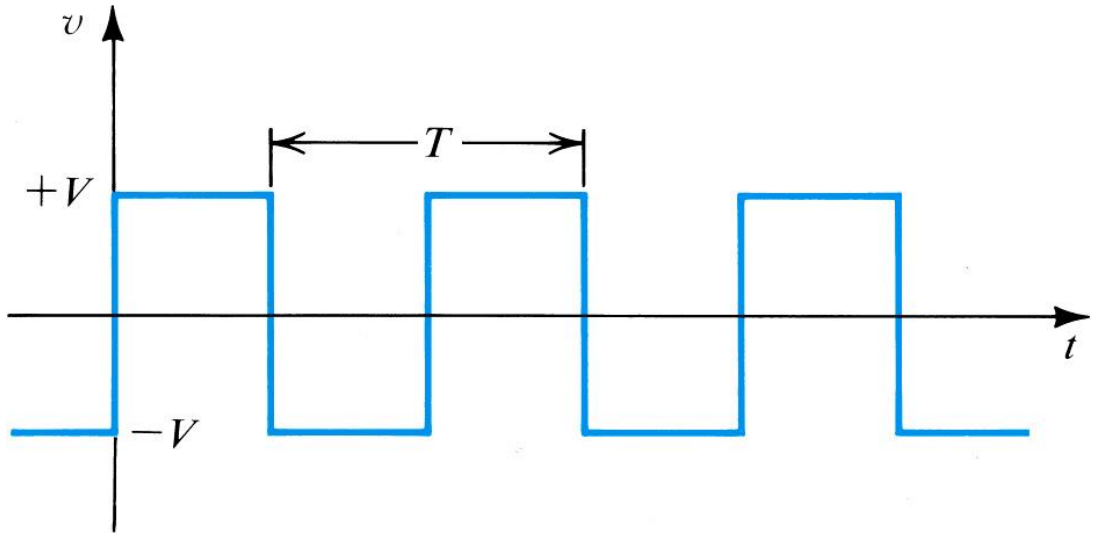
$$\mathbf{f}(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty}\left[a_k \mathbf{cos}(kx) + b_k \mathbf{sin}(kx)\right]$$

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \mathbf{f}(x)\mathbf{cos}(kx)dx,\ n \geq 0$$

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \mathbf{f}(x)\mathbf{sin}(kx)dx,\ n \geq 1$$

What is a Fourier Series? (2)

- **Q:** How does one calculate Fourier Series of square wave below?
  - **A:** See upcoming slides…



Fourier Series Example

step #1: define a_k for the square wave
note that the piece-wise square wave must be divided in two dc functions

$$a_k = \frac{-V_a}{\pi}\underbrace{\int_{-\pi}^{0}\cos(kx)dx}_{\frac{1}{k}\sin(kx)\Big|_{-\pi}^{0}} + \frac{V_a}{\pi}\underbrace{\int_{0}^{\pi}\cos(kx)dx}_{\frac{1}{k}\sin(kx)\Big|_{0}^{\pi}} = 0$$

$$\frac{1}{k}\sin(k0) - \frac{1}{k}\sin(-k\pi) \qquad \frac{1}{k}\sin(k\pi) - \frac{1}{k}\sin(k0)$$

$$0 - 0 \qquad 0 - 0$$

Fourier Series Example

step #2: define b_k for the square wave if **k is even**

$$b_k = \frac{-V_a}{\pi}\underbrace{\int_{-\pi}^{0} V_a \sin(kx)dx}_{-\frac{1}{k}\cos(kx)\Big|_{-\pi}^{0}} + \frac{V_a}{\pi}\underbrace{\int_{0}^{\pi} \sin(kx)dx}_{-\frac{1}{k}\cos(kx)\Big|_{0}^{\pi}}$$

$$-\frac{1}{k}\cos(k0)+\frac{1}{k}\cos(-k\pi) \qquad -\frac{1}{k}\cos(k\pi)+\frac{1}{k}\cos(k0)$$

$$-\frac{1}{k}+\frac{1}{k}(-1)^k \qquad -\frac{1}{k}(-1)^k+\frac{1}{k}$$

$$b_k = \begin{cases} k \text{ is odd} & \dfrac{4V_a}{k\pi} \\ k \text{ is even} & 0 \end{cases}$$

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Fourier Series Example

this series may be truncated because the magnitude of each terms decreases with *k*…

step #3: define Fourier Series

$$\mathbf{f}(x) = \frac{\overset{0}{a_0}}{2} + \sum_{k=1}^{\infty}\left[\overset{0}{a_k}\,\mathbf{cos}(kx) + \overset{\frac{4V_a}{k\pi}}{b_k}\,\mathbf{sin}(kx)\right]$$

$$\mathbf{f}(x) = \frac{4V_a}{\pi}\sum_{k=1}^{\infty}\begin{cases} k \text{ is odd} & \frac{1}{k}\mathbf{sin}(k\omega_0 t) \\ k \text{ is even} & 0 \end{cases} \quad \leftarrow \quad x = \omega_0 t$$

$$\mathbf{f}(x) = \frac{4V_a}{\pi}\left[\mathbf{sin}(\omega_0 t) + \frac{1}{3}\mathbf{sin}(3\omega_0 t) + \frac{1}{5}\mathbf{sin}(5\omega_0 t) + \ldots\right]$$

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Fourier Series Example

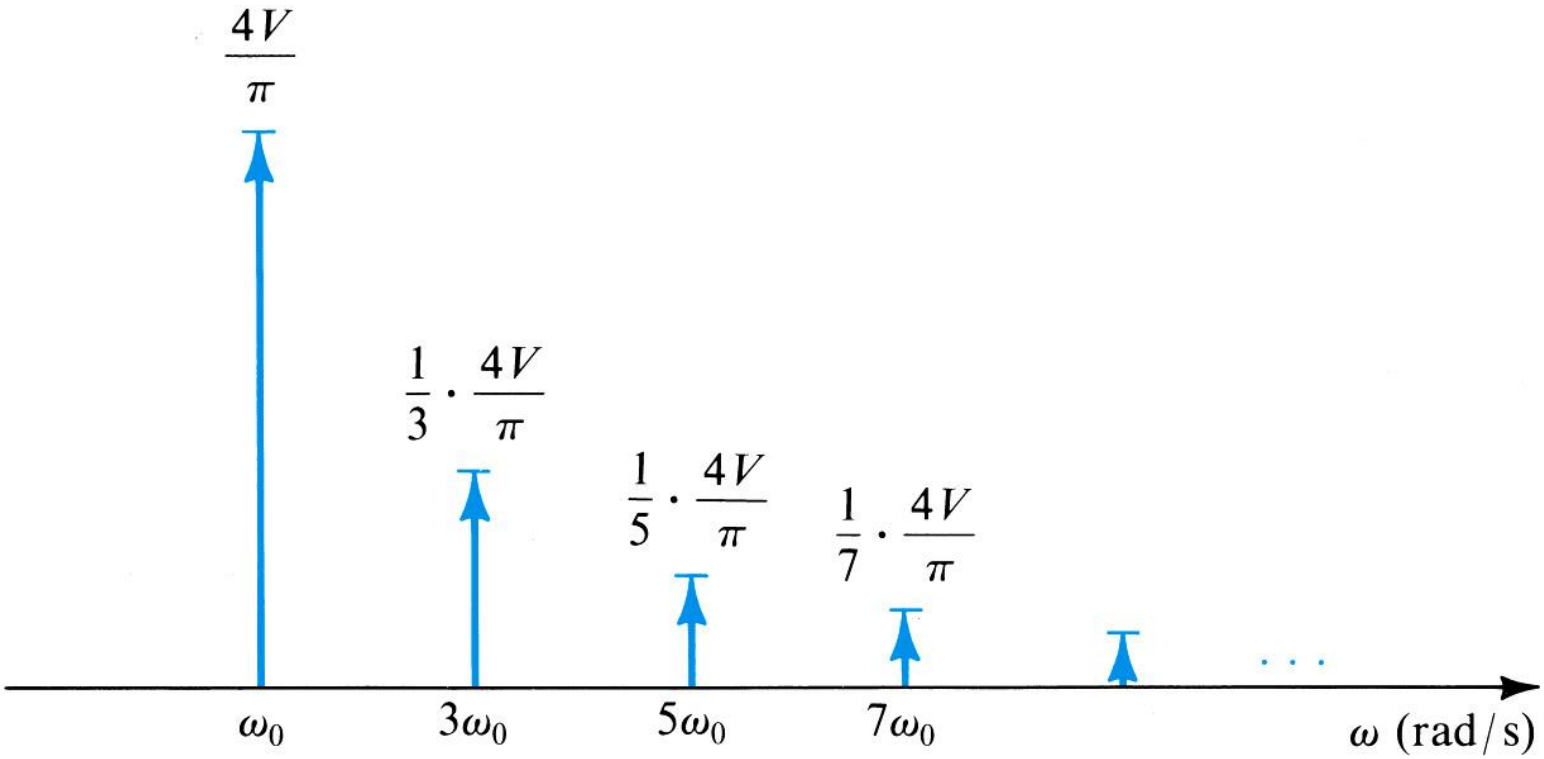


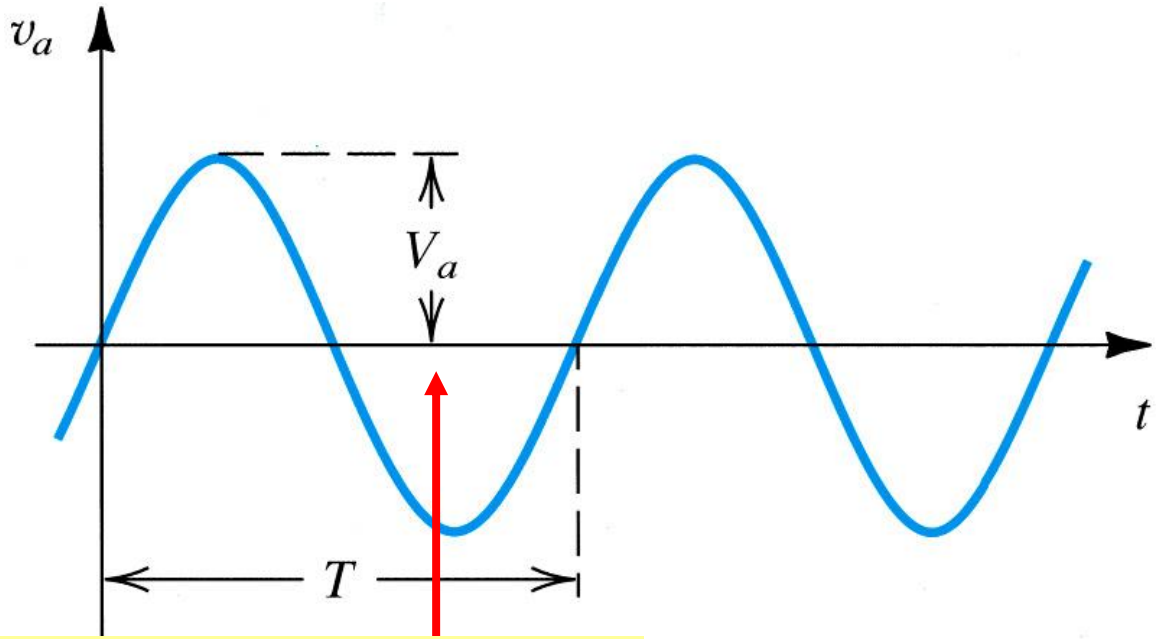
Figure 1.6: The frequency spectrum (also known as the **line spectrum**) of the periodic square wave of Fig. 1.5.

1.2. **Frequency Spectrum of Signals**

- Examine the sinusoidal wave below…

$$\mathbf{v}_a(t) = V_a \sin(\omega t + \theta)$$

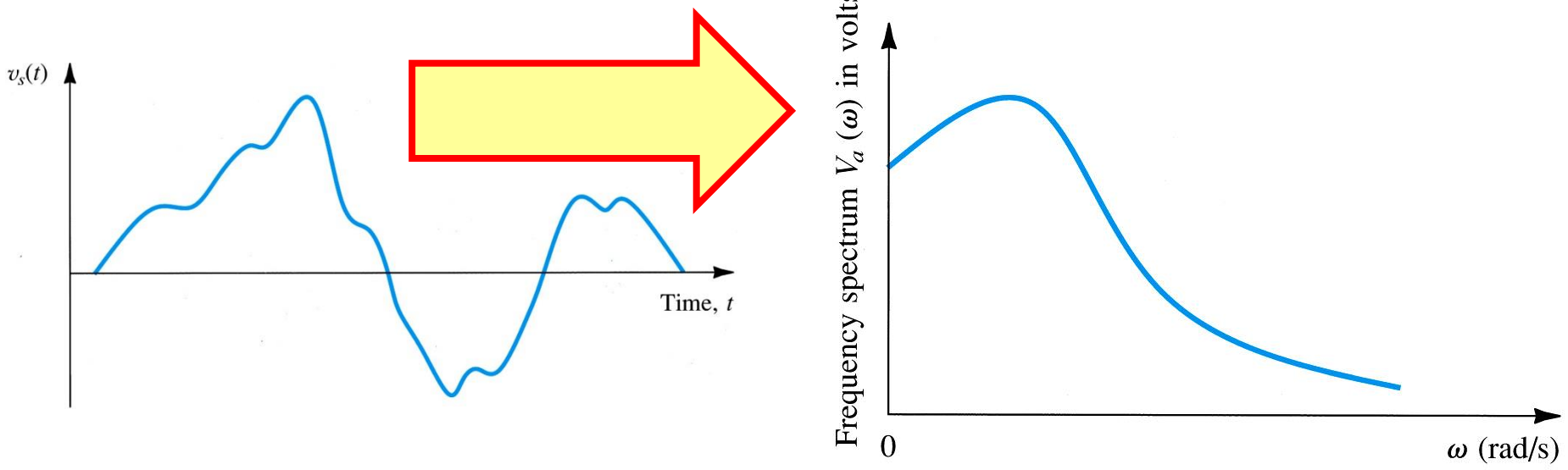
V_a = amplitude in volts
ω = angular frequency in rad/sec
θ = phase shift in rad
t = time in sec



root mean square magnitude =
sine wave amplitude / square root of two

1.2. **Frequency Spectrum of Signals**

- **Q:** Can the Fourier Transform be applied to a non-periodic function of time?
  - **A:** Yes, however (as opposed to a discrete frequency spectrum) it will yield a continuous…



1.3. **Analog and Digital Signals**

- **analog signal** – is continuous with respect to both value and time
- **discrete-time signal** – is continuous with respect to value but sampled at discrete points in time
- **digital signal** – is quantized (applied to values) as well as sampled at discrete points in time

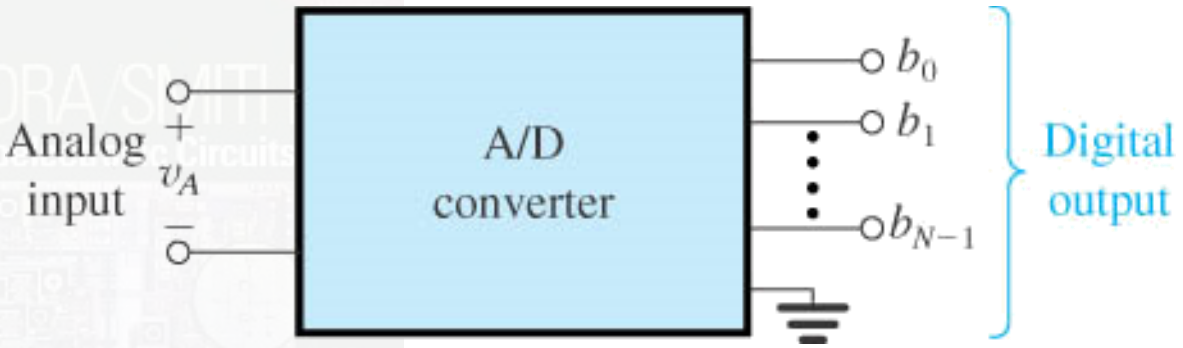
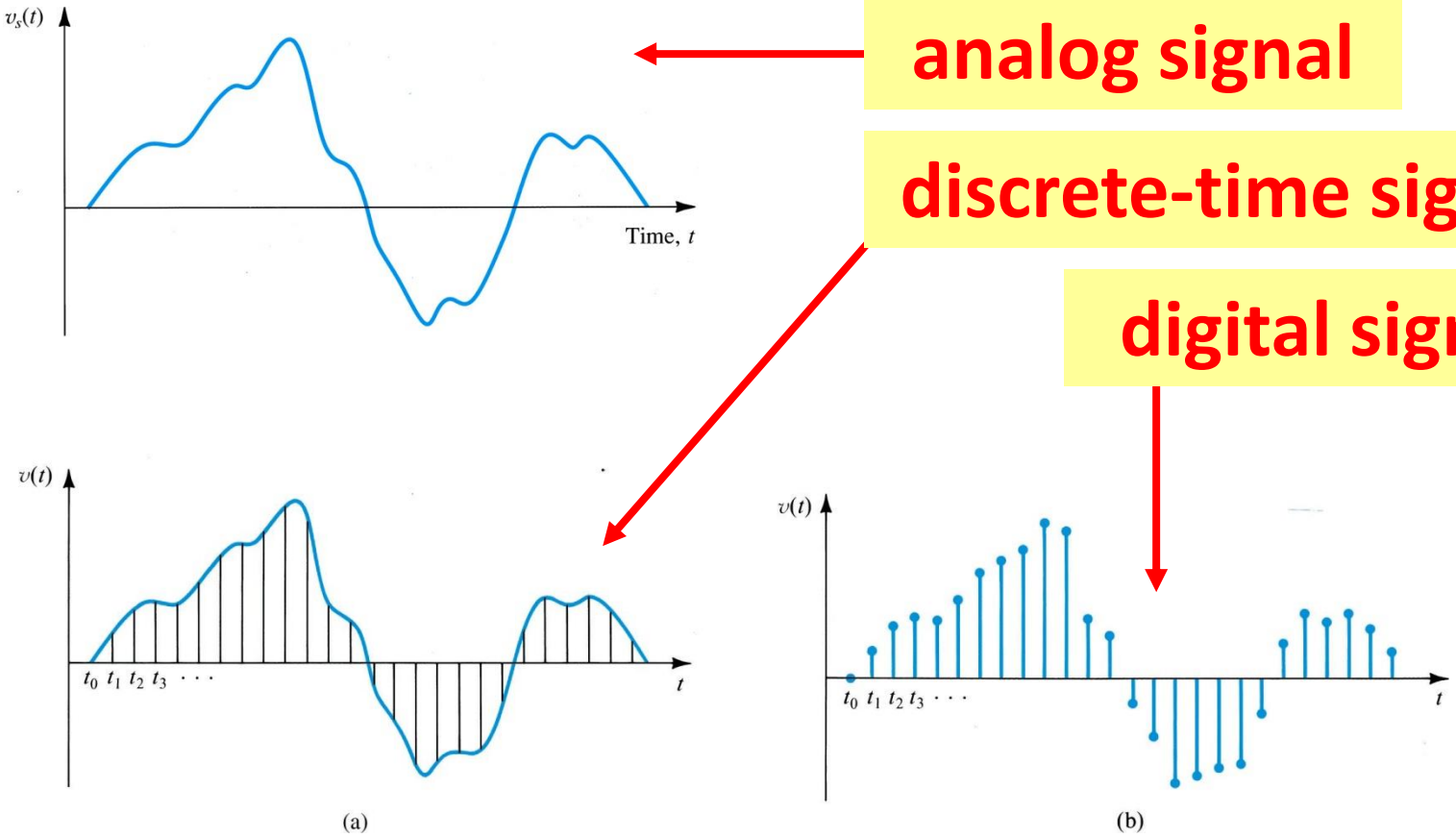


Figure 1.9 Block-diagram representation of the analog-to-digital converter (ADC).

1.3. **Analog and Digital Signals**

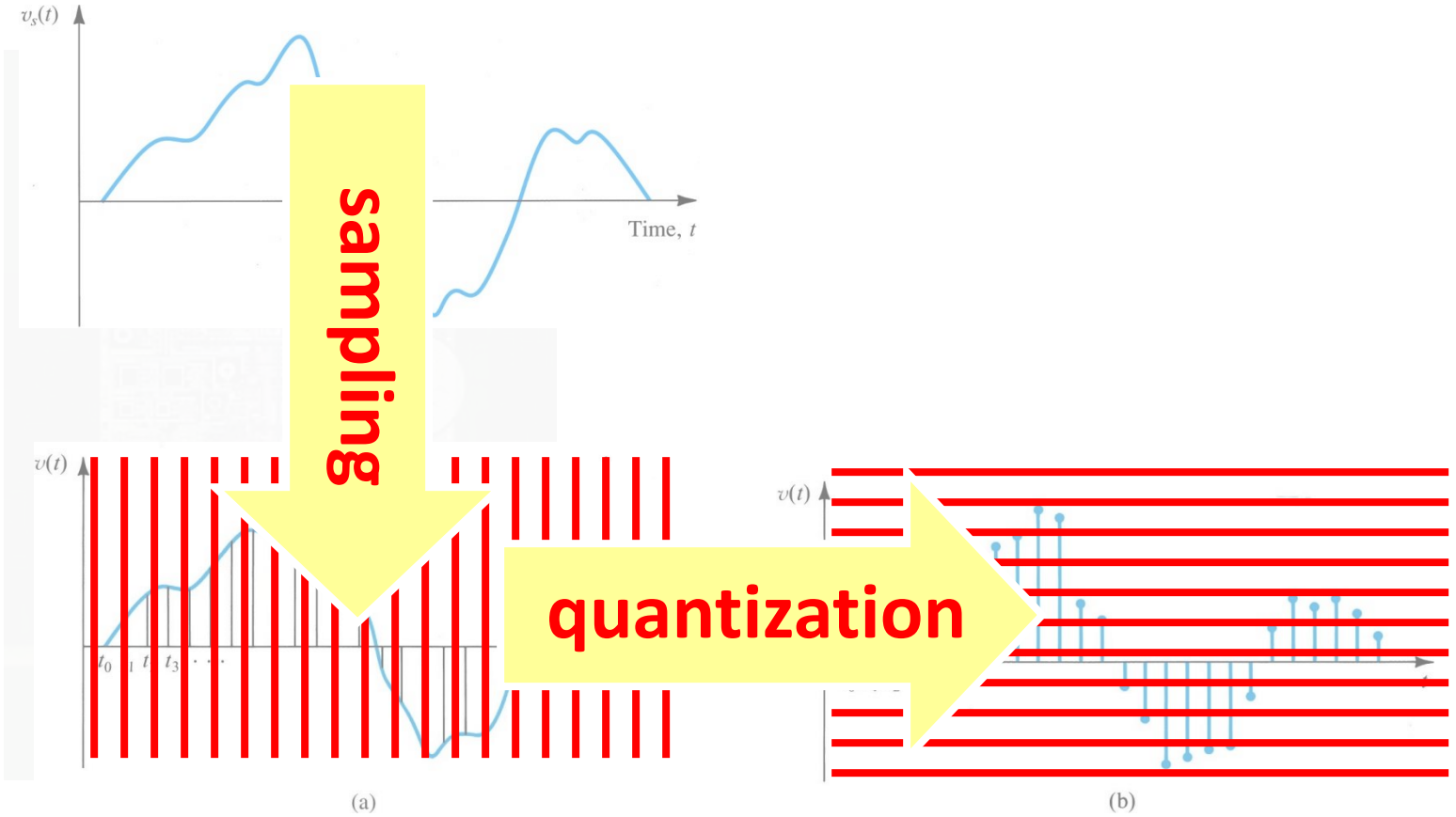


analog signal

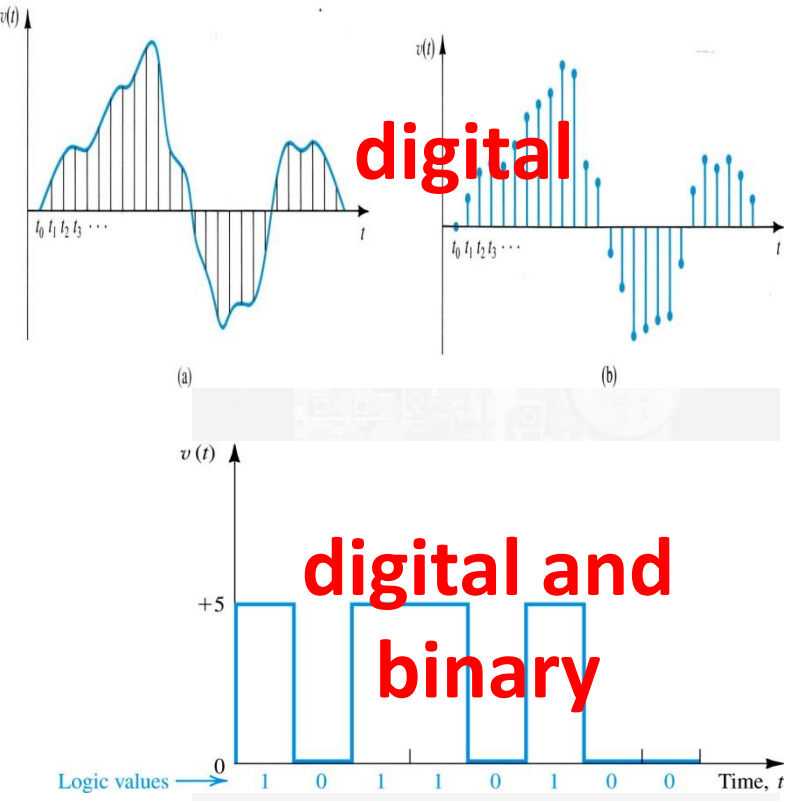
discrete-time signal

digital signal

1.3. **Analog and Digital Signals**



1.3. **Analog and Digital Signals**



- **Q:** Are digital and binary synonymous?
  - **A:** No. The binary number system (base$_2$) is one way to represent digital signals.

base 10 ← base 2

$$y = b_0 2^0 + b_1 2^1 + b_2 2^2 + \ldots$$

LSB

$$\ldots + b_3 2^3 + \ldots b_{n-1} 2^{n-1}$$

MSB

1.4. **Amplifiers**

- **Q:** Why is signal amplification needed?
  - **A:** Because many transducers yield output at low power levels (*mW*)
- **linearity** – is property of an amplifier which ensures a signal is not "altered" from amplification
- **distortion** – is any unintended change in output

1.4.1. Signal Amplification

- **voltage amplifier** – is used to boost voltage levels for increased resolution.
- **power amplifier** – is used to boost current levels for increased "intensity".

output / input relationship for amplifier

$$\mathbf{v}_o(t) = A_v \mathbf{v}_i(t)$$

voltage gain

1.4.2. **Amplifier Circuit Symbol**

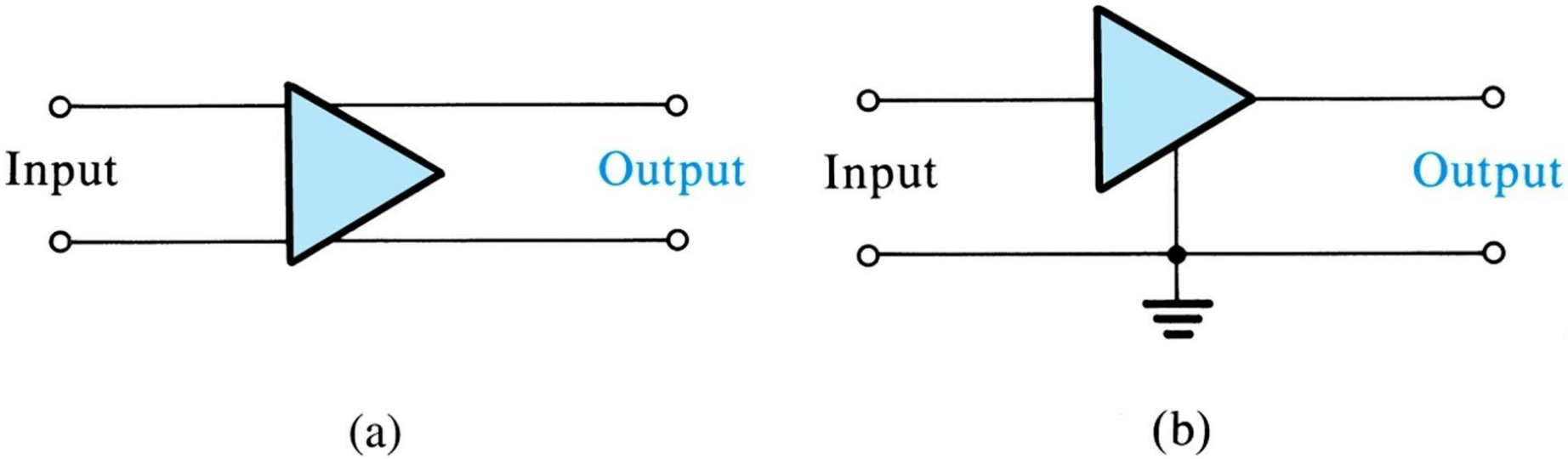


Figure 1.11: (a) Circuit symbol for amplifier. **(b)** An amplifier with a common terminal (ground) between the input and output ports.

1.4.4. **Power and Current Gain**

- **Q:** What is one main difference between an amplifier and transformer? …Because both alter voltage levels.
  - **A:** Amplifier may be used to boost power delivery.

$$power\ gain\ (A_p) = \frac{load\ power\ (P_L)}{input\ power\ (P_I)} = \frac{v_o i_o}{v_i i_i}$$

1.4.5. **Expressing Gain in Decibels**

- **Q:** How may gain be expressed in **decibels**?

$$\text{voltage gain in decibels} = 20\ \mathbf{log}|A_v|\ dB$$

$$\text{current gain in decibels} = 20\ \mathbf{log}|A_i|\ dB$$

$$\text{power gain in decibels} = 10\ \mathbf{log}(A_p)\ dB$$

1.4.6. **Amplifier Power Supply**

- **supplies** – an amplifier has two power supplies
  - V_{CC} is positive, current I_{CC} is drawn
  - V_{EE} is negative, current I_{EE} is drawn
- **power draw** – from these supplies is defined below
  - $P_{dc} = V_{CC} I_{CC} + V_{EE} I_{EE}$

1.4.6. **Amplifier Power Supply**

- **conservation of power** – dictates that power input (P_i) plus that drawn from supply (P_{dc}) is equal to output (P_L) plus that which is dissipated (P_{dis}).
  - $P_i + P_{dc} = P_L + P_{dissapated}$
- **efficiency** – is the ratio of power output to input.
  - efficiency = $P_L / (P_i + P_{dc})$

1.4.6. **Amplifier Power Supply**

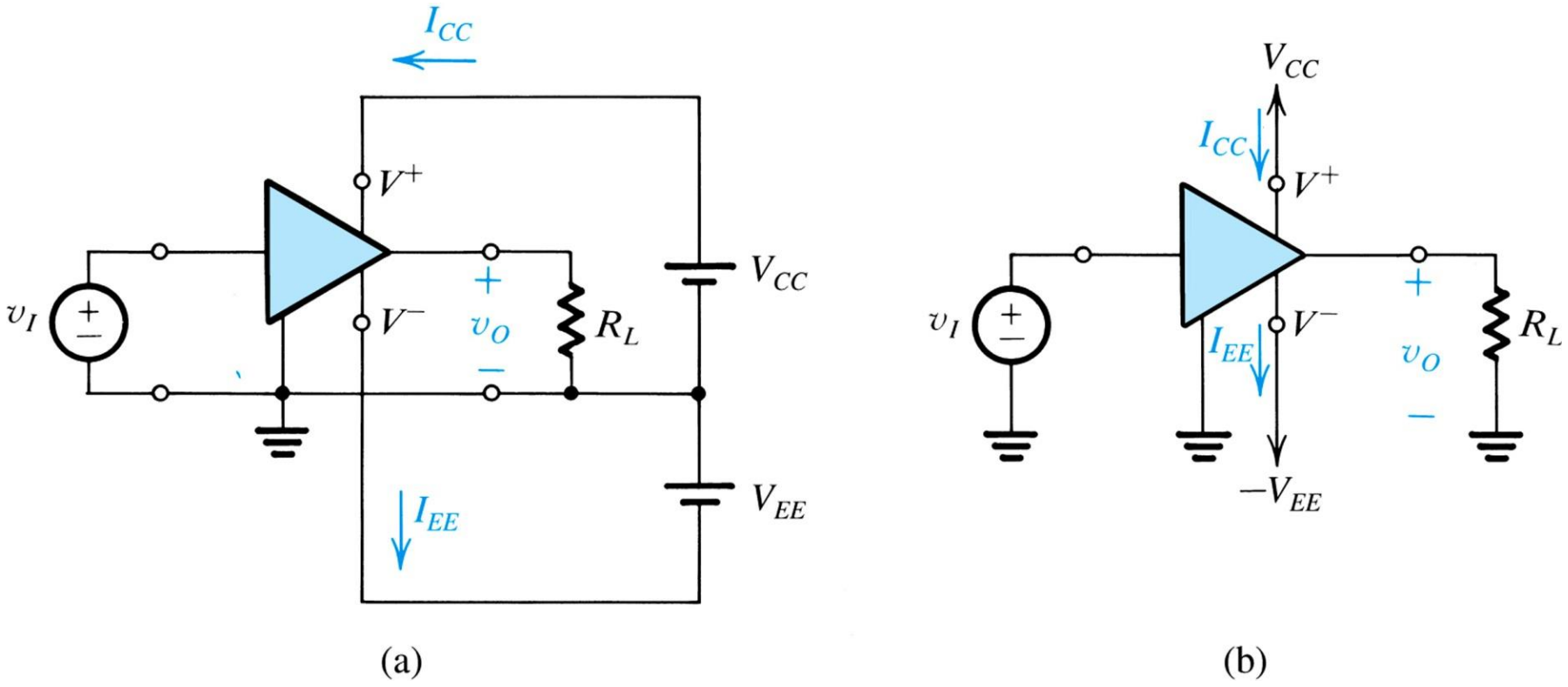


Figure 1.13: An amplifier that requires two dc supplies (shown as batteries) for operation.

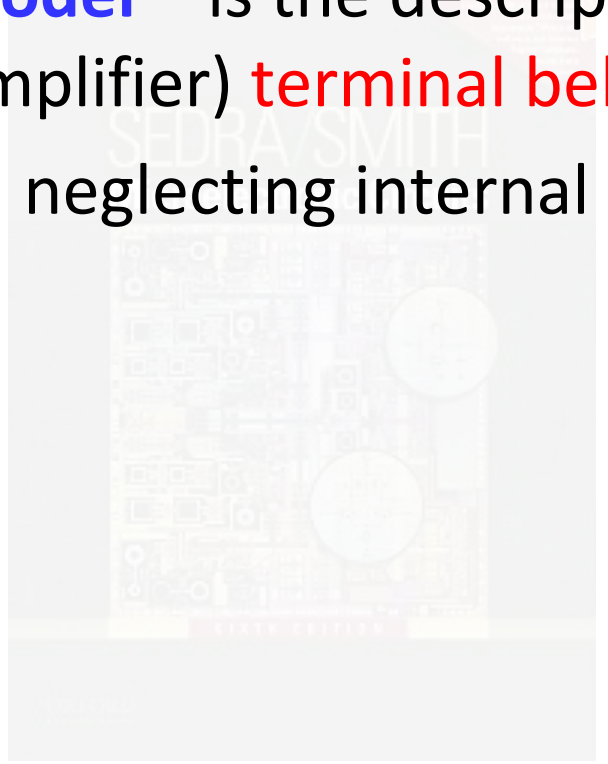
1.4.7. **Amplifier Saturation**

- **limited linear range** – practically, amplifier operation is linear over a limited input range.
- **saturation** – beyond this range, saturation occurs.
  - output remains constant as input varies

$$\underbrace{\frac{L_{minus}}{A_v} \leq v_i \leq \frac{L_{plus}}{A_v}}_{\text{or...}\ L_{minus} \leq v_o \leq L_{plus}}$$

1.5. **Circuit Models for Amplifiers**

- **model** – is the description of component's (e.g. amplifier) terminal behavior
  - neglecting internal operation / transistor design



1.5.1. Voltage Amplifiers

model of amplifier input terminals

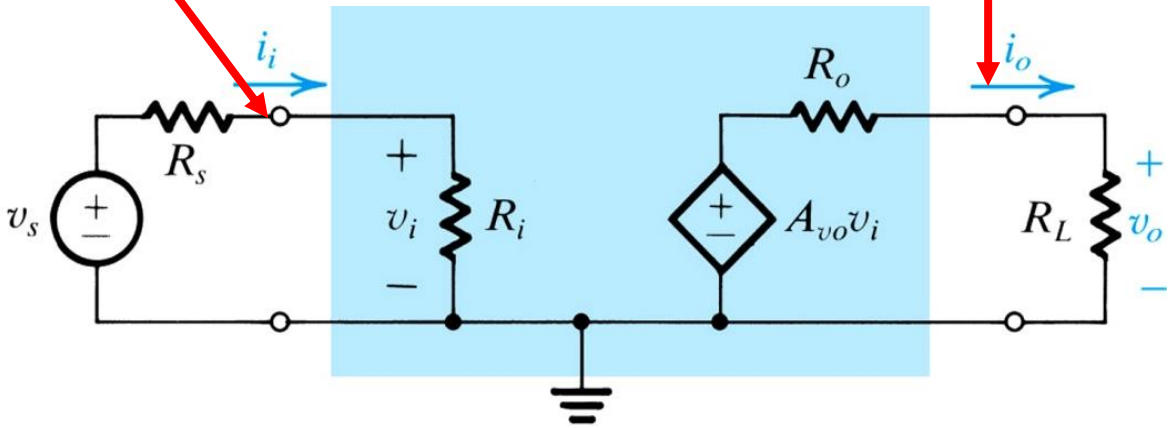
$$\text{input voltage} = v_i = (v_s)\underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}}$$

source volt.

model of amplifier output terminals

$$\text{output voltage} = v_o = (A_{vo} v_i)\underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}}$$

open-ckt output voltage



(b)

1.5.1. **Voltage Amplifiers**

- **Q:** How can one model the amplifier behavior from previous slide?
  - **A:** Model which is function of: v_s, A_{vo}, R_i, R_s, R_o, R_L

$$v_o = \underbrace{\left(A_{vo} \underbrace{(v_s)}_{\text{source volt.}} \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}} \right)}_{\text{open-ckt output voltage}} \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}} = A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

1.5.1. Voltage Amplifiers

- **Q:** What is one "problem" with this behavior?
  - **A:** Gain (ratio of v_o and v_s) is not constant, and dependent on input and load resistance.

$$v_o = \underbrace{A_{vo}\,(v_s)}_{\text{source volt.}}\ \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}}\ \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}} = A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

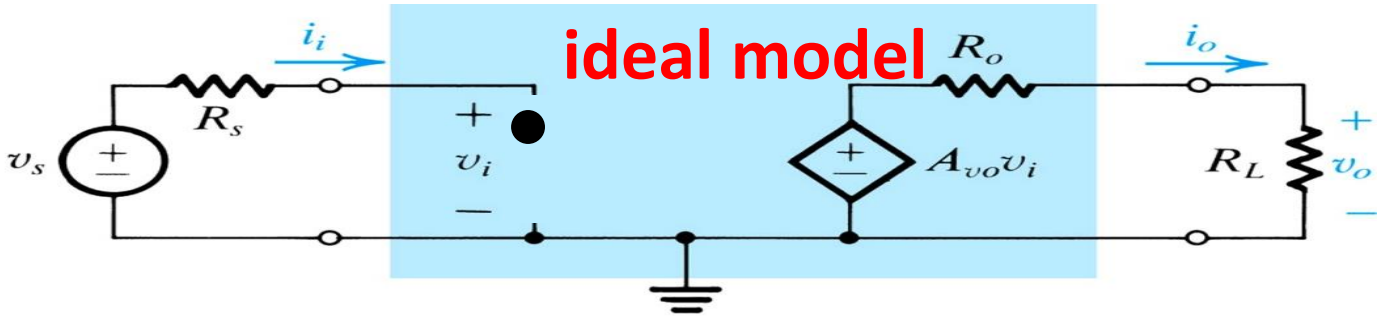
The **ideal amplifier model** neglects this nonlinearity.

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1.5.1. Voltage Amplifiers

- **ideal amplifier model** – is function of v_s and A_{vo} only!!
  - It is assumed that $R_o << R_L$…
  - It is assumed that $R_i << R_s$…

non-ideal model



key characteristics of ideal voltage amplifier model = high input impedance, low output impedance

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1.5.1. **Voltage Amplifiers**

- **ideal amplifier model** – is function of v_s and A_{vo} only!!
  - It is assumed that $R_o << R_L$…
  - It is assumed that $R_i << R_s$…

$$v_o = \underbrace{A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}}_{\text{non-ideal model}} = \underbrace{A_{vo} v_s}_{\text{ideal model}}$$

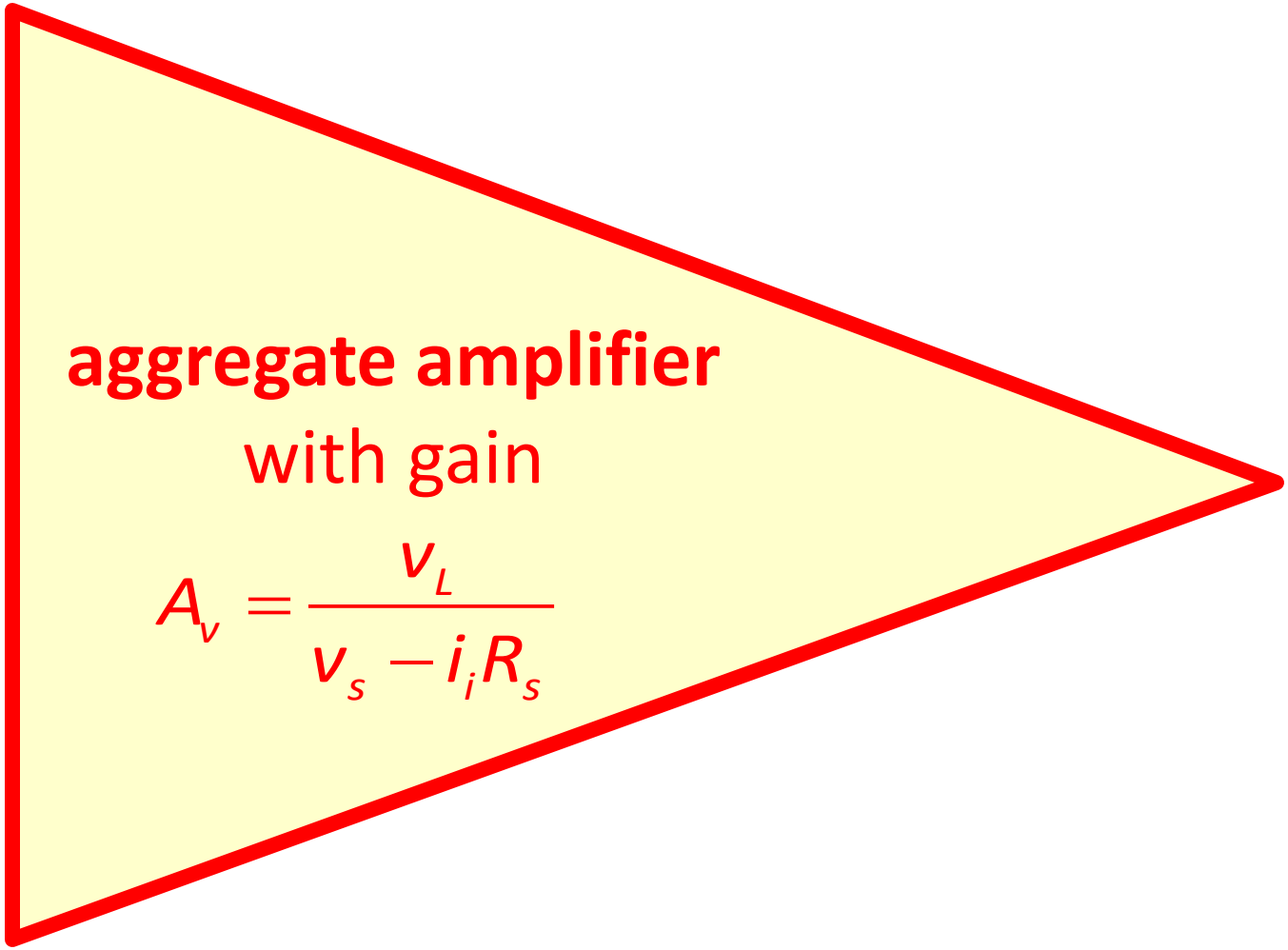
key characteristics of ideal voltage amplifier model = source resistance R_S and load resistance R_L have no effect on gain

1.5.2. **Cascaded Amplifiers**

- In real life, an amplifier is not ideal and will not have infinite input impedance or zero output impedance.
- **Cascading** of amplifiers, however, may be used to emphasize desirable characteristics.
  - first amplifier – high R_i, medium R_o
  - last amplifier – medium R_i, low R_o
  - aggregate – high R_i, low R_o

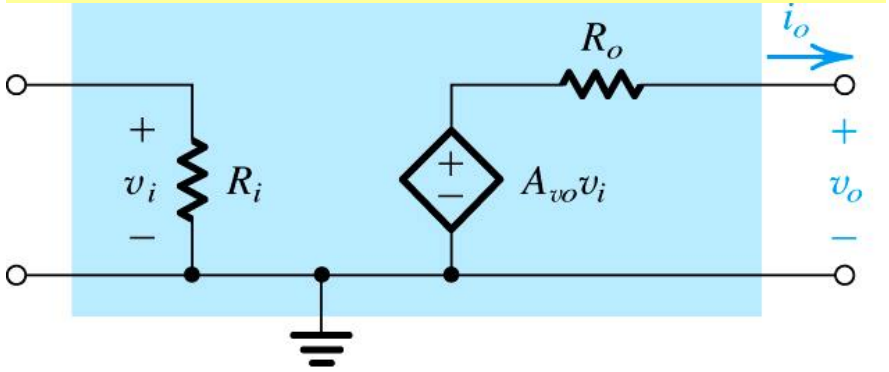
Example 1.3: Cascaded Amplifier Configurations

- Examine system of cascaded amplifiers on next slide.
- **Q(a):** What is overall voltage gain?
- **Q(b):** What is overall current gain?
- **Q(c):** What is overall power gain?

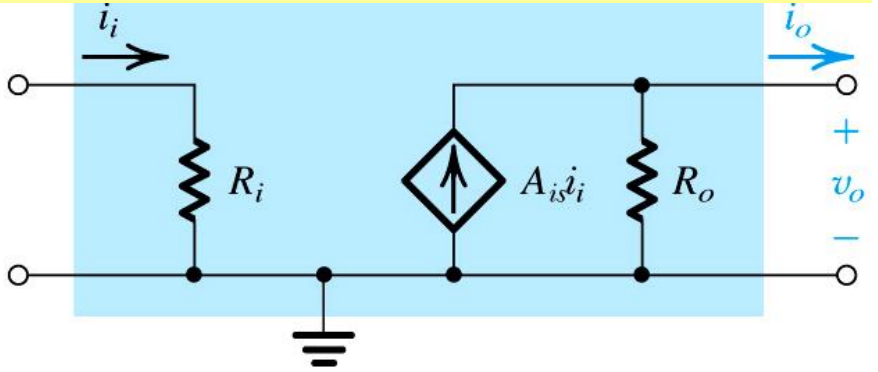

aggregate amplifier
with gain
$A_v = \frac{v_L}{v_s - i_i R_s}$

1.5.3. **Other Amplifier Types**

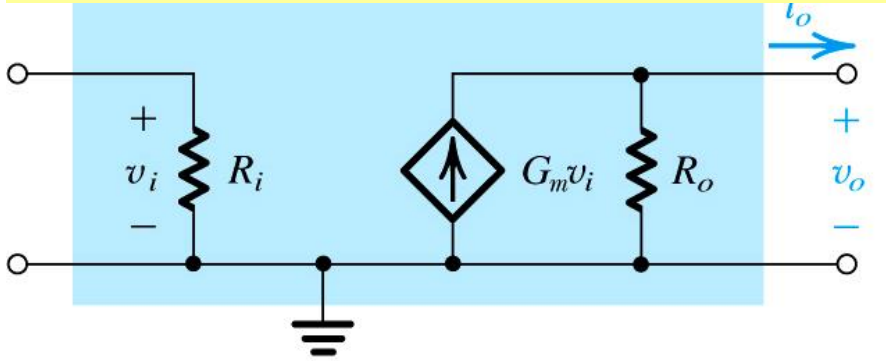
voltage amplifier



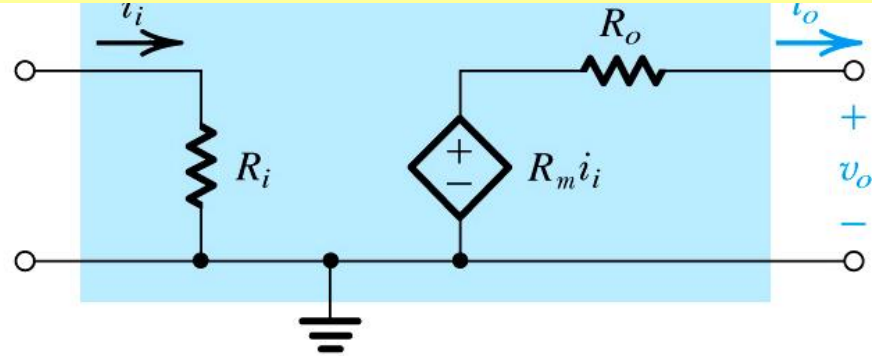
current amplifier



transconductance amp.

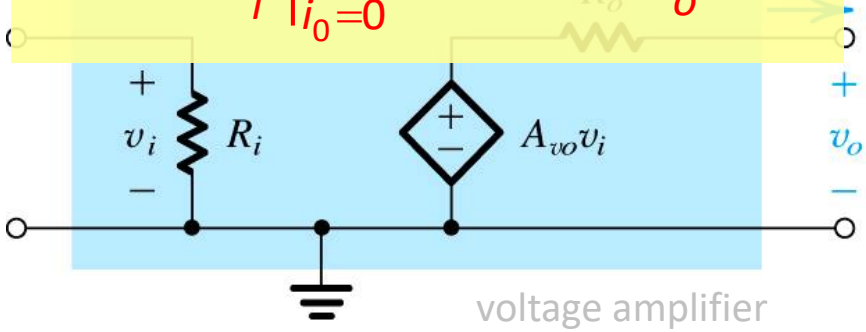


transresistance amp.



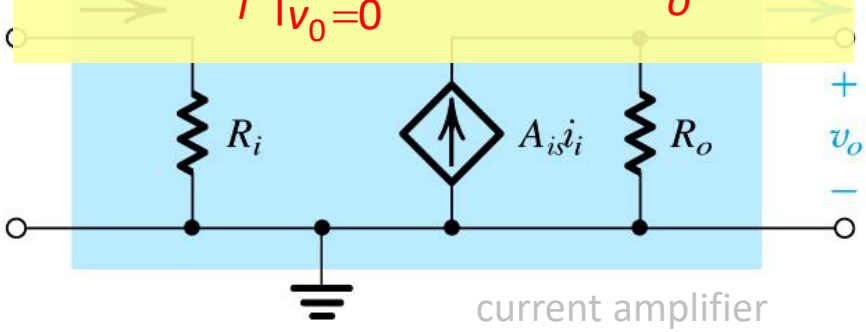
1.5.3. **Other Amplifier Types**

$$A_{v0} = \left.\frac{v_0}{v_i}\right|_{i_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = 0$$



voltage amplifier

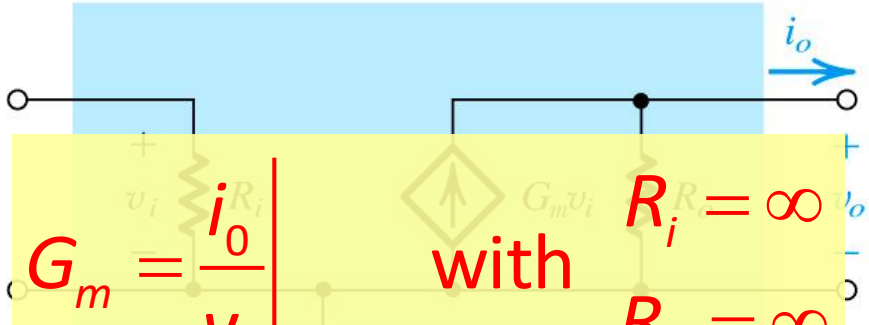
$$A_{v0} = \left.\frac{i_0}{i_i}\right|_{v_0=0} \quad \text{with} \quad R_i = 0, \; R_o = \infty$$



current amplifier

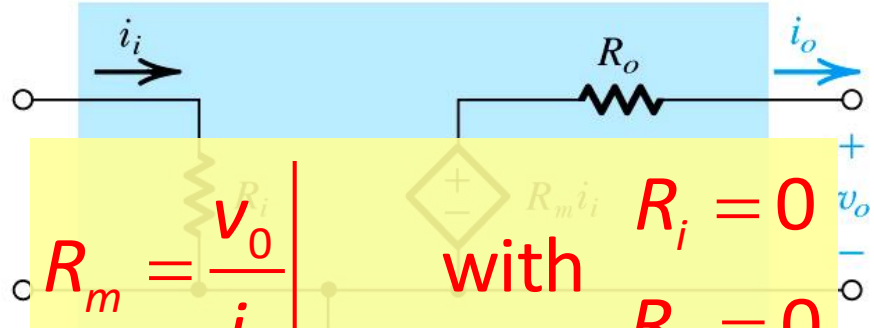
transconductance amplifier

$$G_m = \left.\frac{i_0}{v_i}\right|_{v_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = \infty$$



transresistance amplifier

$$R_m = \left.\frac{v_0}{i_i}\right|_{i_0=0} \quad \text{with} \quad R_i = 0, \; R_o = 0$$



1.5.4. Relationship Between Four Amp Models

- **interchangeability** – although these four types exist, any of the four may be used to model any amplifier
  - they are related through A_{vo} (open circuit gain)

current to voltage amplifier

transcond. to voltage amplifier

transres. to voltage amplifier

$$A_{vo} = A_{is}\left(\frac{R_o}{R_i}\right) = G_m R_o = \frac{R_m}{R_i}$$

1.5.5. Determining R_i and R_o

- **Q:** How can one calculate input resistance from terminal behavior?
  - **A:** Observe v_i and i_i, calculate via $R_i = v_i / i_i$
- **Q:** How can one calculate output resistance from terminal behavior?
  - **A:**
    - Remove source voltage (such that $v_i = i_i = 0$)
    - Apply voltage to output (v_x)
    - Measure negative output current ($-i_o$)
    - Calculate via $R_o = -v_x / i_o$

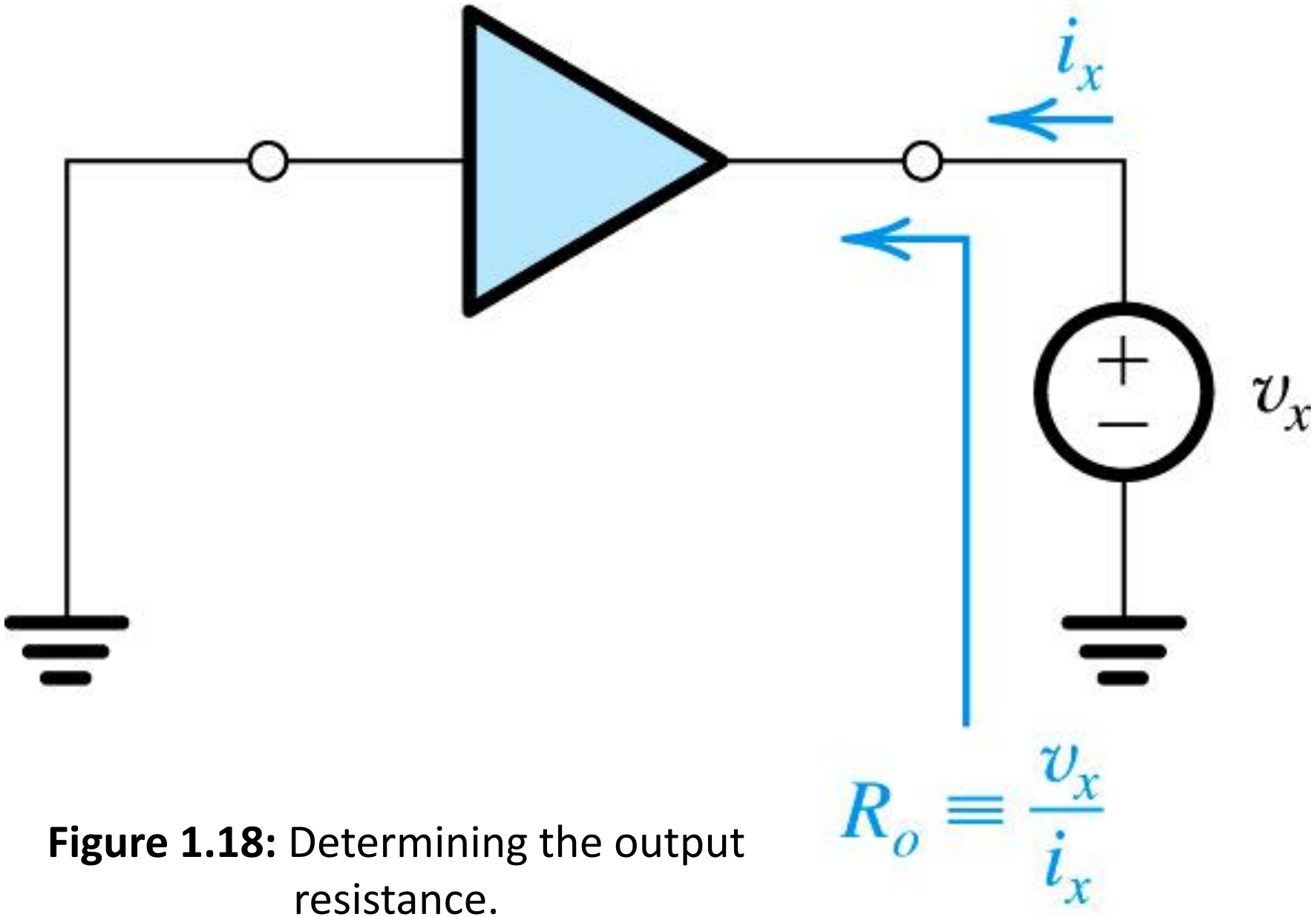


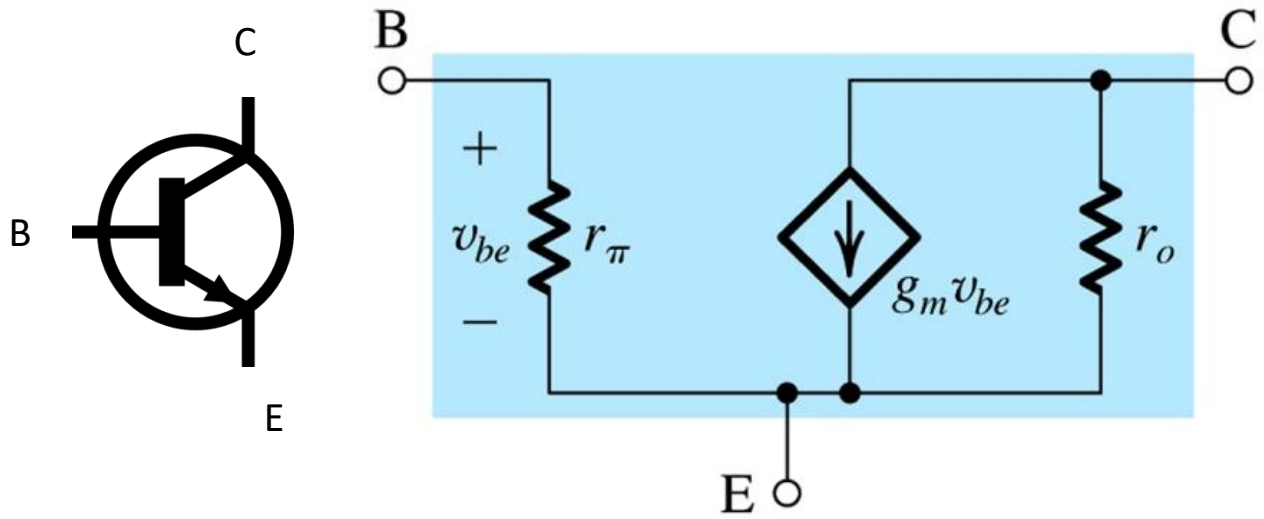
Figure 1.18: Determining the output resistance.

1.5.6. **Unilateral Models**

- **unilateral model** – is one in which signal flows only from input to output (not reverse)
  - However, most practical amplifiers will exhibit some reverse transmission…

Example 1.4: Common-Emitter Circuit

- Examine the **bipolar junction transistor** (BJT).
  - three-terminal device
  - when powered up with dc source and operated with small signals, may be modeled by linear circuit below.



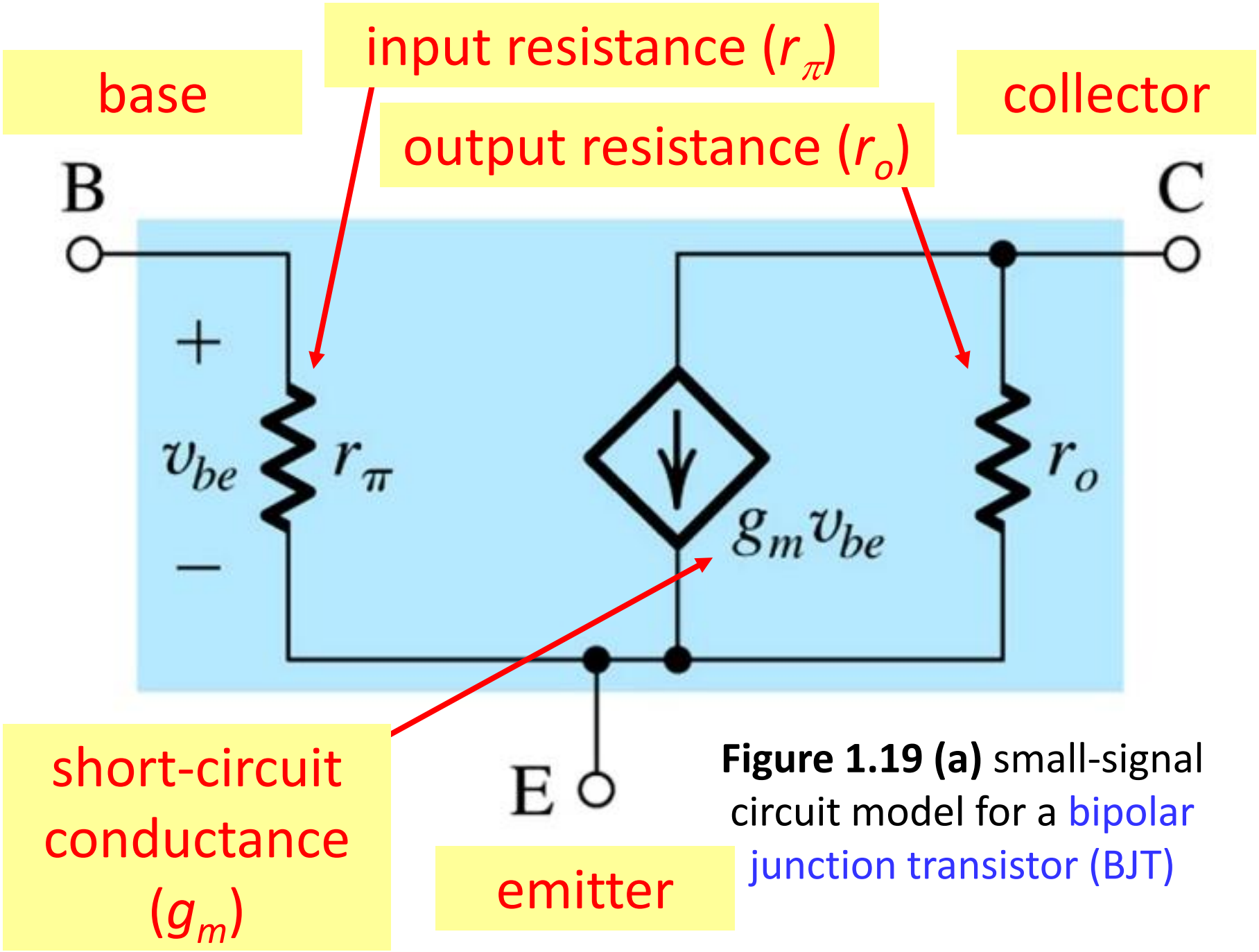


Figure 1.19 (a) small-signal circuit model for a bipolar junction transistor (BJT)

Example 1.4: Common-Emitter Circuit

- **Q(a):** Derive an expression for the voltage gain v_o / v_i of common-emitter circuit with:
  - $R_s = 5 kohm$
  - $r_\pi = 2.5 kohm$
  - $g_m = 40 mA/V$
  - $r_o = 100 kohm$
  - $R_L = 5 kohm$

input and output share common terminal

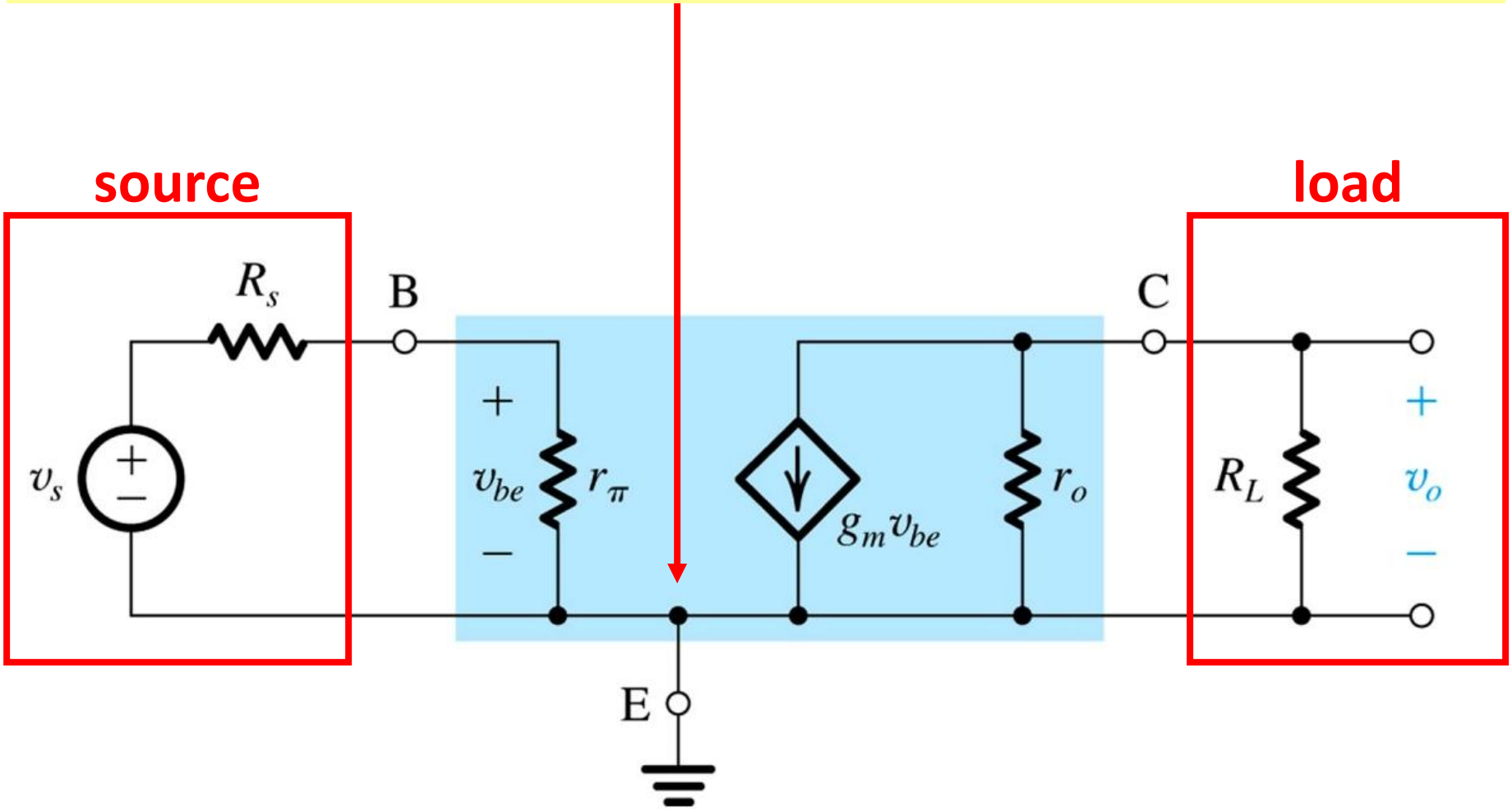


Figure 1.19(b): The BJT connected as an amplifier with the emitter as a common terminal between input and output (called a common-emitter amplifier).

1.6.1. **Measuring the Amplifier Frequency Response**

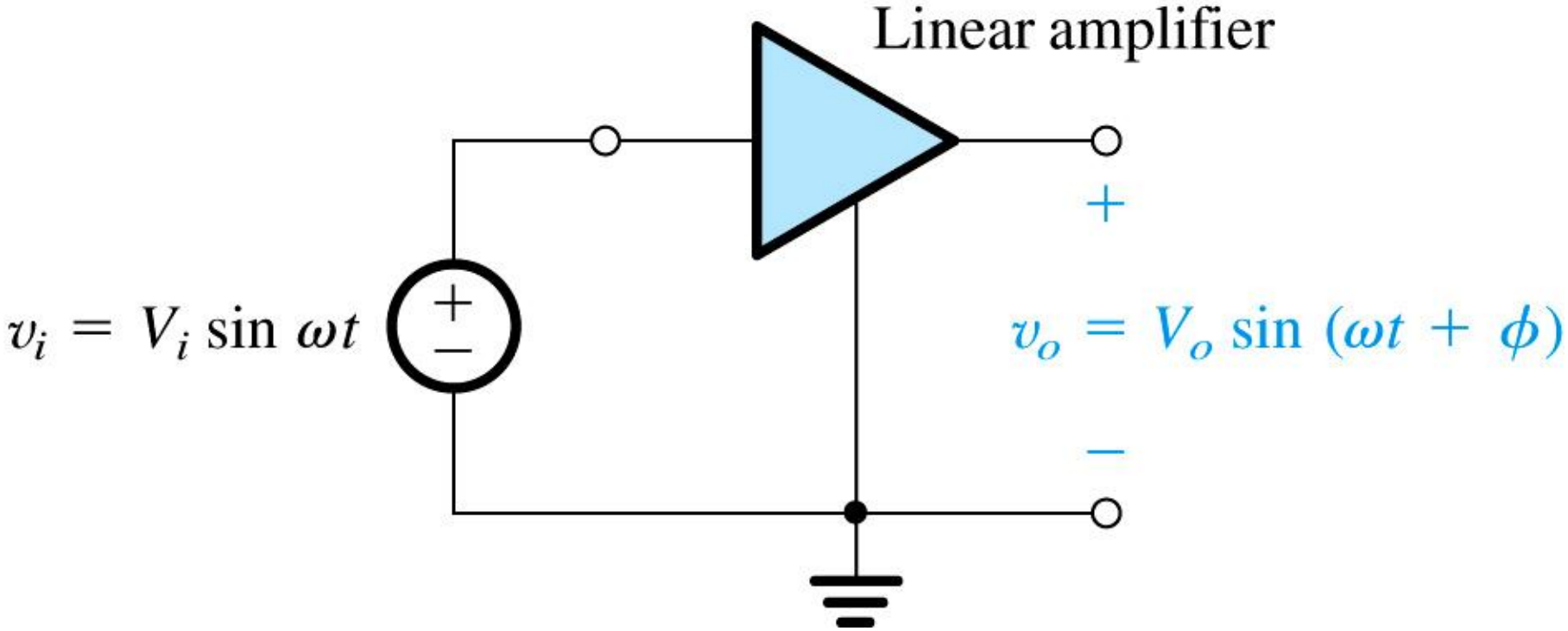
- **Q:** How does one examine frequency response?
  - **A:** By applying sine-wave input of amplitude V_i and frequency ω.
- **Q:** Why?
  - **A:** Because, although its amplitude and phase may change, its shape and frequency will not.

this characteristic of sine wave applied to linear circuit is unique

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1.6.1: Measuring the Amplifier Frequency

input and output are similar for linear amplifier



1.6.1. Measuring the Amplifier Frequency Response

- **amplifier transfer function** (**T**) – describes the input-output relationship of an amplifier – or other device – with respect to various parameters, including frequency of input applied.
  - It is a complex value, often defined in terms of magnitude and phase shift.

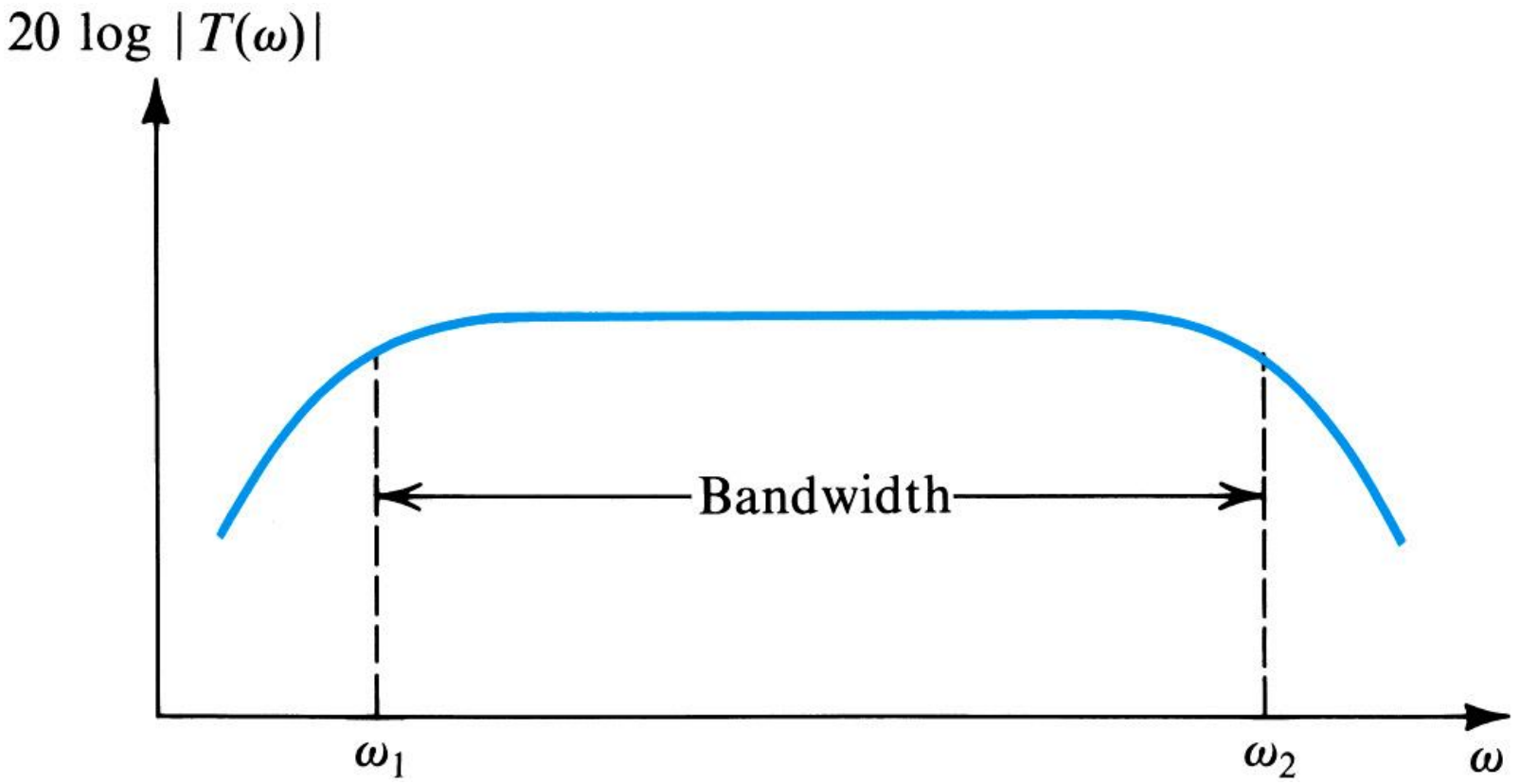
$$\underbrace{\left|\mathbf{T}(\omega)\right| = \frac{V_o}{V_i}}_{\text{magnitude gain}} \quad \text{and} \quad \underbrace{\angle\mathbf{T}(\omega) = \phi}_{\text{phase shift}}$$

1.6.2. **Amplifier Bandwidth**

- **Q:** What is **bandwidth** of a device?
  - **A:** The range of frequencies over which its magnitude response is constant (within 3dB).
- **Q:** For an amplifier, what is main bandwidth concern?
  - **A:** That the bandwidth extends beyond range of frequencies it is expected to amplify.

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1.6.2. Amplifier Bandwidth



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1.6.4. **Single Time-Constant Networks**

- **single time–constant (STC) network** – is composed of (or may be reduced to) one reactive component and one resistance.
  - **low pass filter** – attenuates output at high frequencies, allow low to pass
  - **high pass filter** – attenuates output at low frequencies, allow high to pass
- **time constant (τ)** – describes the length of time required for a network transient to settle from step change ($\tau = L / R = RC$)

1.6.4. **Single Time-Constant Networks**

- low pass filter (left)
- high pass filter (right)

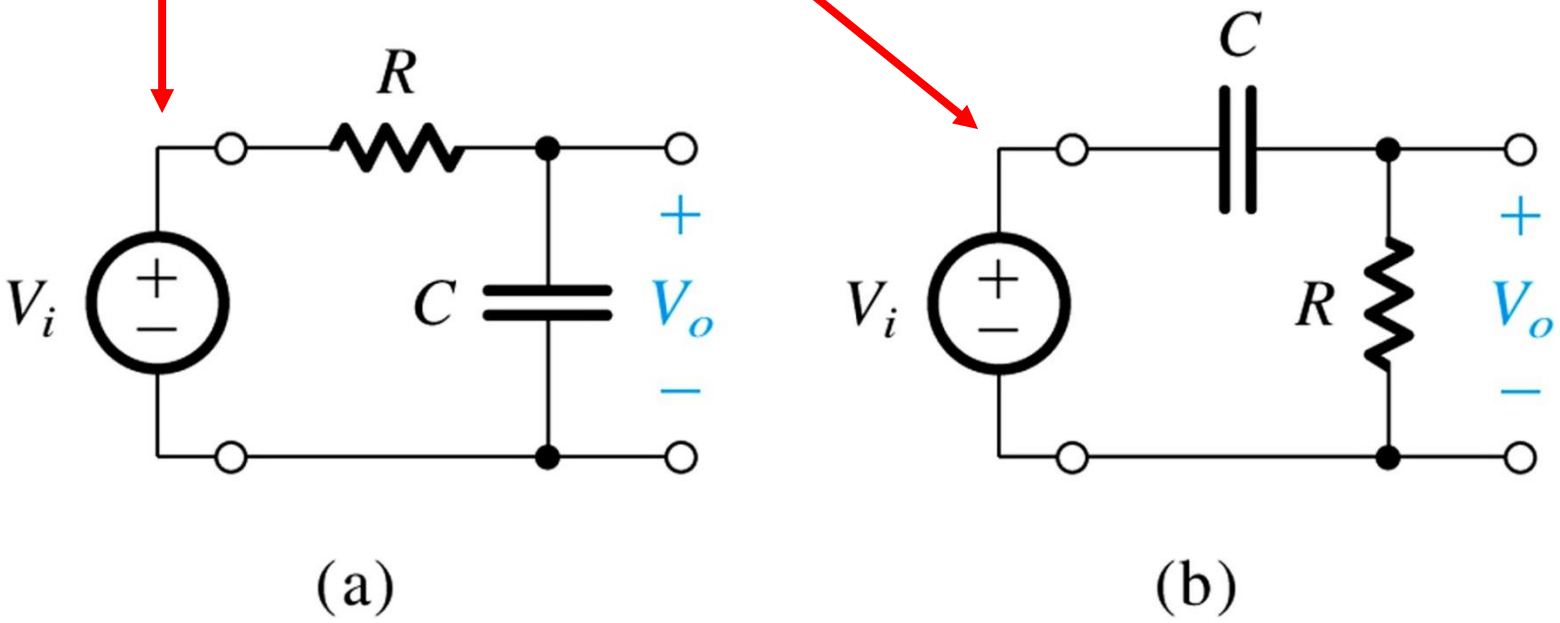


Figure 1.22: Two examples of STC networks: **(a)** a low-pass network and **(b)** a high-pass network.

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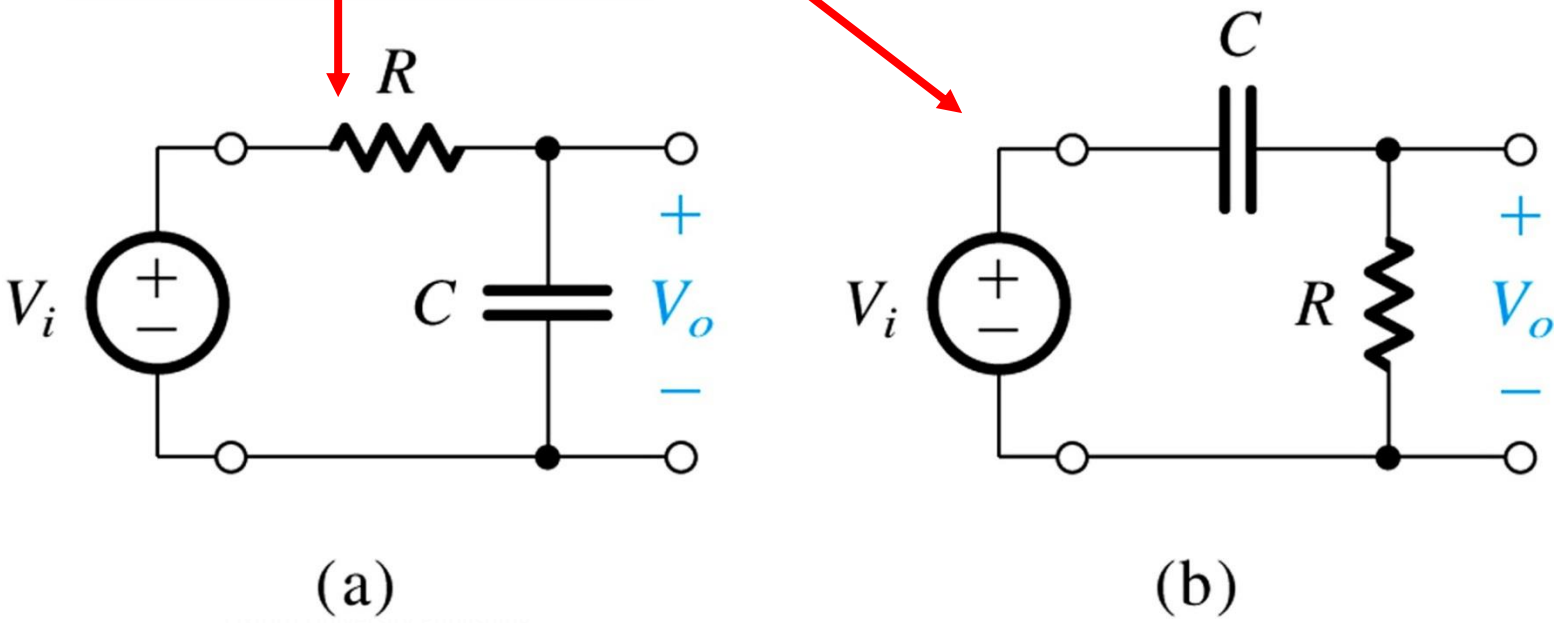
1.6.4. Single Time-Constant Networks

low-pass:

$$\frac{v_o}{v_i} = \frac{Z_o}{Z_i + Z_o} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \propto k\frac{1}{\omega}$$

high-pass:

$$\frac{v_o}{v_i} = \frac{Z_o}{Z_i + Z_o} = \frac{R}{R + \frac{1}{j\omega C}} \propto k\omega$$



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1.6.4. Single Time-Constant Networks

Figure 1.2 : Characteristics of Various STC

	low - pass	**high - pass**
transfer function	$\frac{K}{1+(s/\omega_0)}$	$\frac{Ks}{1+\omega_0}$
transfer function (for physical freq.)	$\frac{K}{1+j(\omega/\omega_0)}$	$\frac{K}{1-j(\omega_0/\omega)}$
magnitude response	$\frac{\lvert K\rvert}{\sqrt{1+j(\omega/\omega_0)^2}}$	$\frac{\lvert K\rvert}{\sqrt{1+j(\omega_0/\omega)^2}}$
phase response	$-\tan(\omega/\omega_0)$	$\tan(\omega_0/\omega)$
transmission at $\omega=0$	K	0
transmission at $\omega=\infty$	0	K
3*db* **Frequency**	$\omega_0=\frac{1}{\tau}$	same
Bode Plots	refer to	next slide

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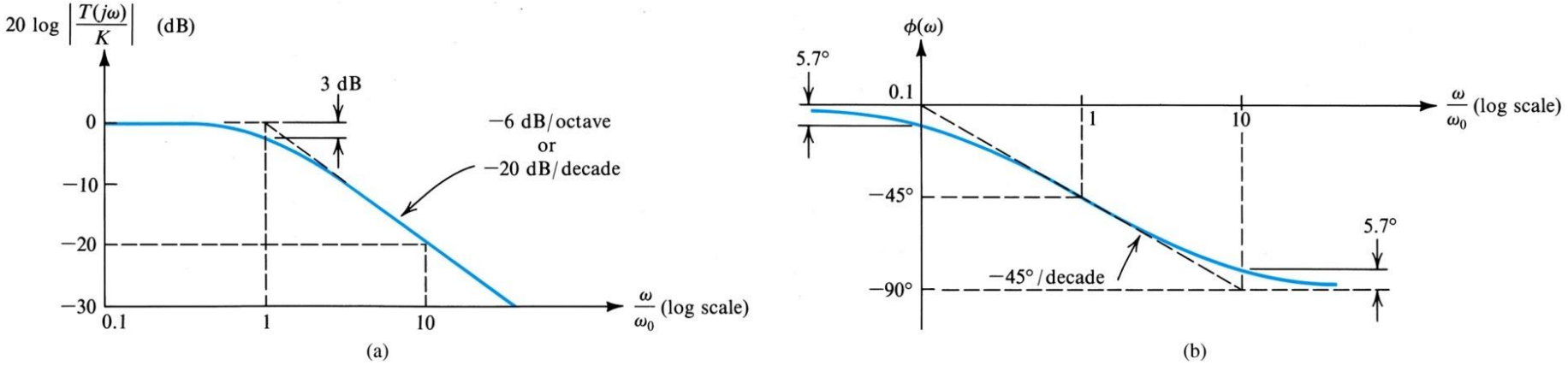
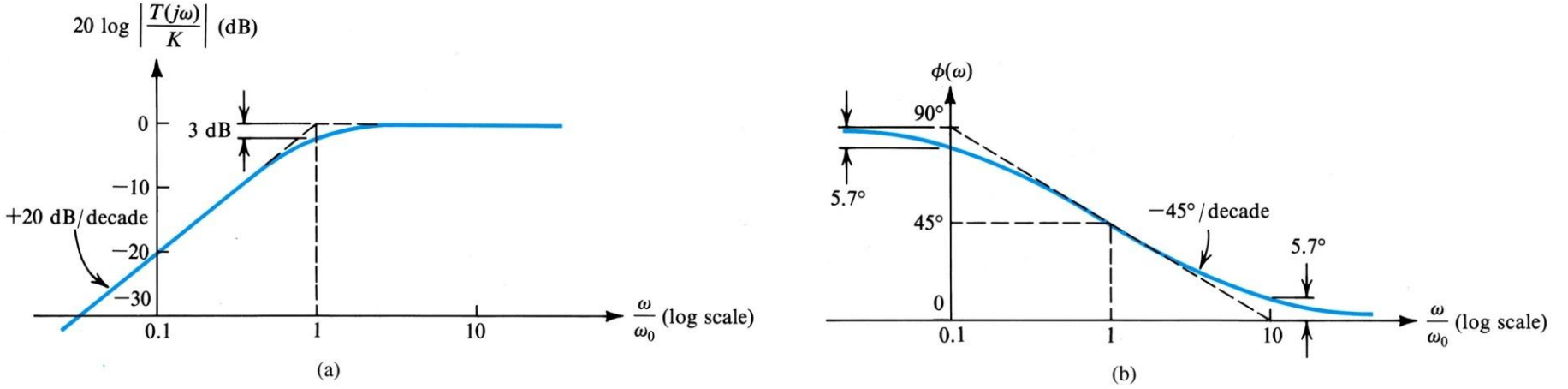
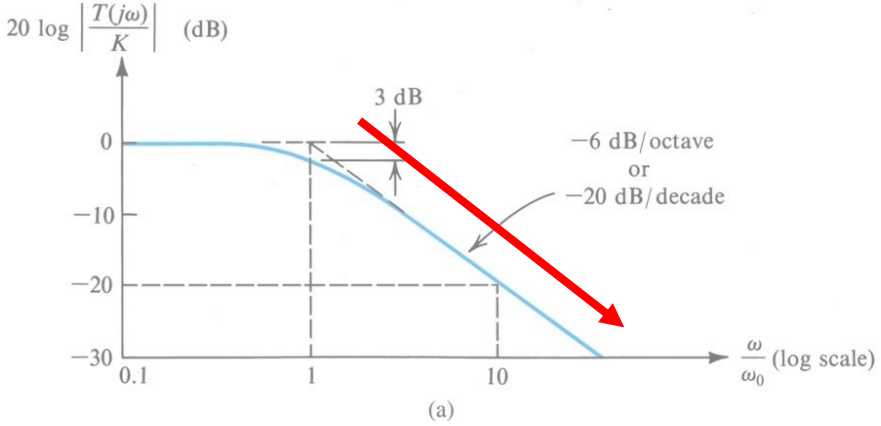
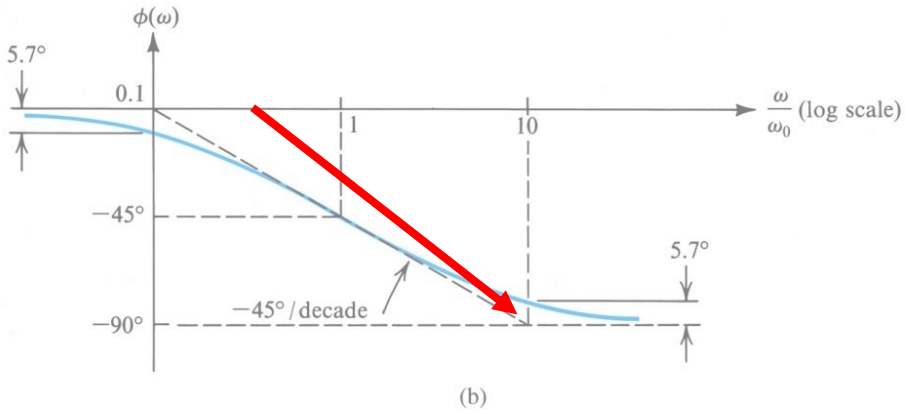


Figure: Low-Pass Filter Magnitude (top-left) and Phase (top-right) Responses as well as High-Pass Filter (bottom-left) and Phase (bottom-right) Responses







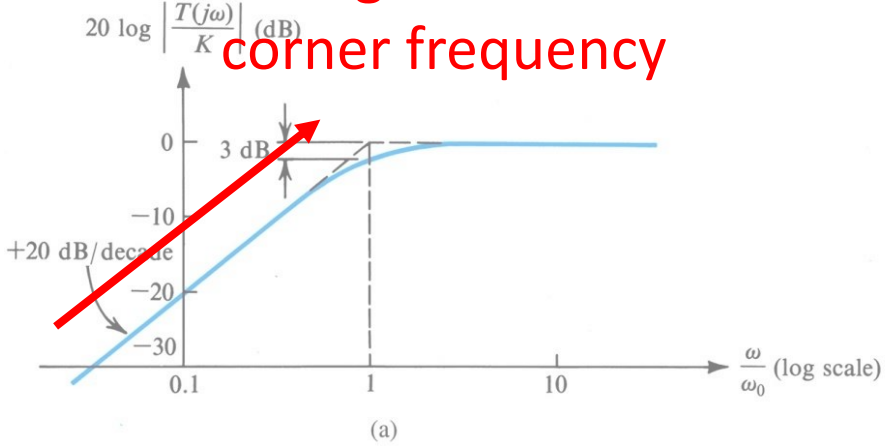
-20dB/decade drop, beginning from maximum gain at corner frequency

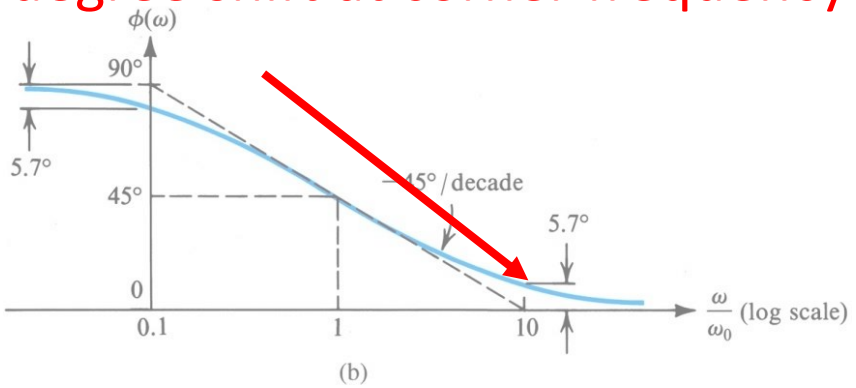
-45 degrees/decade drop, moving outward from -45 degree shift at corner frequency

Figure: Low-Filter Magnitude (top-left) and Phase (top-right) Responses as well as High-Pass Filter (bottom-left) and Phase (bottom-right) Responses

+20dB/decade incline, until maximum gain is reached at corner frequency

-45 degrees/decade drop, moving outward from +45 degree shift at corner frequency





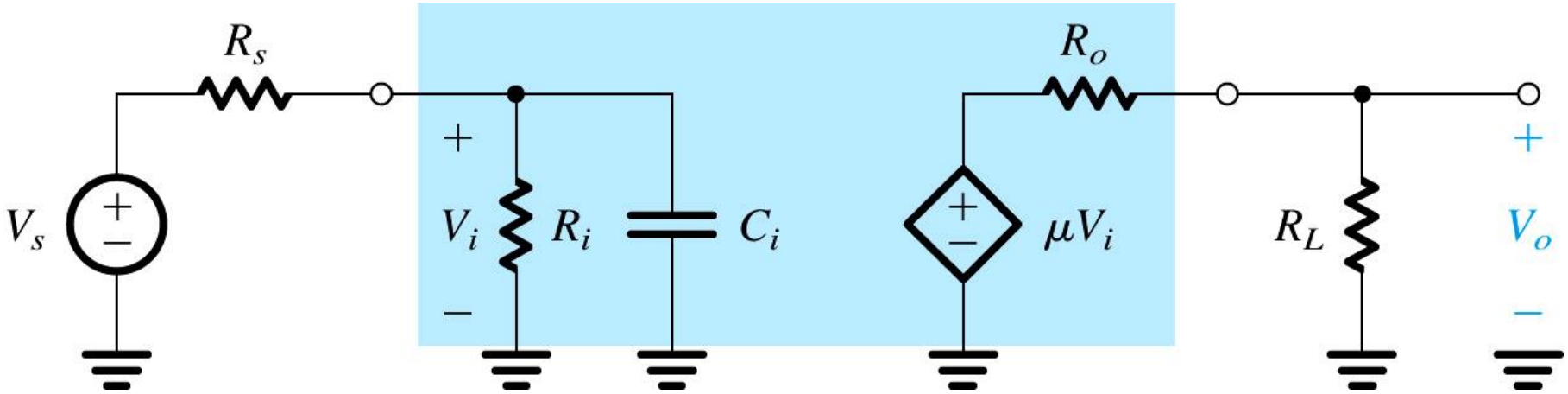
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Example 1.5:
Voltage Amplifier

- Examine voltage amplifier with:
  - input resistance (R_i)
  - input capacitance (C_i)
  - gain factor (μ)
  - output resistance (R_o)
- **Q(a):** Derive an expression for the amplifier voltage gain V_o / V_s as a function of frequency. From this, find expressions for the dc gain and 3dB frequency.

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Example 1.5:
Voltage Amplifier



Example 1.5:
Voltage Amplifier

- **Q(b):** What is unity-gain frequency? How is it calculated?
  - **A:** Gain = 0*dB*
  - **A:** It is known that the gain of a low-pass filter drops at 20dB per decade beginning at ω_0. Therefore unity gain will occur two decades past ω_0 (40*dB* – 20*dB* – 20*dB*).
- **Q(c):** Find $v_o(t)$ for each of the following input: v_s = 0.1**sin**($10^2 t$), v_s = 0.1**sin**($10^5 t$)

1.6.5. **Classification of Amps Based on Frequency Response**

- **internal capacitances** – cause the falloff of gain at high frequencies
  - like those seen in previous example
- **coupling capacitors** – cause the falloff of gain at low frequencies
  - are placed in between amplifier stages
  - generally chosen to be large

1.6.5. **Classification of Amps Based on Frequency Response**

- **directly coupled / dc amplifiers** – allow passage of low frequencies
- **capacitively coupled amplifiers** – allow passage of high frequencies
- **tuned amplifiers** – allow passage of a “band” of frequencies

Conclusion

- An electrical signal source can be represented in either Thevenin form (a voltage source v_s in series with source resistance R_s) or the Norton form (a current source i_s in parallel with resistance R_s). The Thevenin voltage v_s is the open-circuit voltage between the source terminals. The Norton current i_s is equal to the short-circuit current between the source terminals. For the two representations to be equivalent, v_s and $R_s i_s$ must be equal.
- A signal can be represented either by its waveform vs time or as the sum of sinusoids. The latter representation is known as the frequency spectrum of the signal.

Conclusion (2)

- The sine-wave signal is completely characterized by its peak value (or rms value which is the peak / $2^{1/2}$), frequency (ω in rad/s of f in Hz; $\omega = 2\pi f$ and $f = 1/T$, where T is the period is seconds), and phase with respect to an arbitrary reference time.
- Analog signals have magnitudes that can assume any value. Electronic circuits that process analog signals are called analog circuits. Sampling the magnitude of an analog signal at discrete instants of time and representing each signal sample by a number results in a digital signal. Digital signals are processed by digital circuits.

Conclusion (3)

- The simplest digital signals are obtained when the binary number system is used. An individual digital signal then assumes one of only two possible values: low and high (e.g. 0V and 5V) corresponding to logic 0 and logic 1.
- An analog-to-digital converter (ADC) provides at its output the digits of the binary number representing the analog signal sample applied to its input. The output digital signal can then be processed using digital circuits.
- A transfer characteristic, v_o vs. v_i, of a linear amplifier is a straight line with a slope equal to the voltage gain.

Conclusion (4)

- Amplifiers increase the signal power and thus require dc power supplies for their operation.
- The amplifier voltage gain can be expressed as a ratio A_v in V/V or in decibels, $20\log|A_v|$ in dB.
- Depending on the signal to be amplified (voltage or current) and on the desired form of output signal (voltage or current) there are four basic amplifier types: voltage, current, transconductance, and transresistance. A given amplifier may be modeled by any of these configurations, in which case their parameters are related by (1.14) through (1.16) in the text.

Conclusion (5)

- The sinusoid is the only signal whose waveform is unchanged through a linear circuit. Sinusoidal signals are used to measure the frequency response of amplifiers.
- The transfer function $\mathbf{T}(s) = \mathbf{V}_o(s)/\mathbf{V}_i(s)$ of a voltage amplifier may be determined from circuit analysis. Substituting $s = j\omega$ gives $\mathbf{T}(j\omega)$ whose magnitude ($|\mathbf{T}(j\omega)|$ is the magnitude response and $\phi(\omega)$ is the phase response.
- Amplifiers are classified according to the shape of their frequency response.

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Conclusion (6)

- Single-time-constant (STC) networks are those networks that are composed of, or may be reduced to, one reactive component (*L* or *C*) and one resistance. The time constant (τ) is *L*/*R* or *RC*.
- STC networks can be classified into two categories: low-pass (LP) and high-pass (HP). LP network pass dc and low-frequencies while attenuating high-frequencies. The opposite is true for HP.
- The gain of an LP (HP) STC circuit drops by 3*dB* below the zero-frequency (infinite-frequency) value at a frequency $\omega_0 = 1/\tau$. At high-frequencies (low-frequencies) the gain falls of at a rate of 6*dB*/octave or 20*dB*/decade.
  - Refer to Table 1.2. on page 34 and Figs. 1.23 and 1.24. Further details are provided in Appendix E.